Decorated Discriminant Normalization and Stable Moduli

Repository working draft

July 22, 2026

Abstract

For the weighted marked-root Keller maps attached to a normalized seed H, we study the normalization of the discriminant curve of the pencil $H(W) - sW + t$. Its zeroth relative Fitting ideal is $(H''(r))$ with full multiplicity. We retain the saturated off-diagonal node-pairing scheme, the conductor morphism, the point at infinity, and the scheme-valued boundary marks. On a nonempty ordinary boundary-clean open, the resulting stable invariant is generically etale of degree $N-2$ onto a normalized image of dimension $N-3$. Consequently degree-N weighted maps contain an $(N-3)$-dimensional family of stable classes. Full-cover faithfulness belongs to this package but is isolated below as a repair target and is not used in the decorated-normalization degree or stable-moduli proof.

1 Setup and scope

Let k be a characteristic-zero field and let $H \in k[W]$ be a normalized admissible degree-N seed:

$$H(0) = H'(0) = H(1) = 0, \qquad H'(1) = -1.$$

The weighted construction associates to H a polynomial Keller map F_H and the marked inverse equation

$$E_H(W; s, t) = H(W) - sW + t = 0.$$

Its finite incidence normalization is equipped with the maximal open on which the rational source reconstruction is regular. The stable-normalization functoriality theorem in the degreewise-multiplicity paper identifies the intrinsic discriminant vertex, its finite normalization morphism, and its boundary marks under polynomial left–right equivalence and after adjoining identity variables. Those facts are the functorial input used here.

The decorated invariant deliberately forgets part of the regular- reconstruction open. Its generic degree is therefore a rerooting degree, not a faithfulness assertion. The separate full-cover argument later in this draft requires repair before it is promoted to an input for other results; none of the Fitting, node, conductor, generic-degree, or stable-moduli results depends on it.

2 The scheme-theoretic decorated normalization

For a weighted primitive H, let D_H be the reduced discriminant of $H(W) - sW + t$ and let

$$\nu_H : \mathbb{A}^1_r \longrightarrow D_H, \qquad r \longmapsto (H'(r), rH'(r) - H(r)) \tag{6.1}$$

be its affine normalization. Write $R_H = k[H'(r), rH'(r) - H(r)] \subset k[r]$.

Proposition 2.1 (full Fitting divisor)**.** *There is an equality of ideal sheaves on the normalization*

$$\operatorname{Fitt}_0 \Omega_{\widetilde{D}_H/D_H} = (H''(r)). \tag{6.2}$$

This ideal, including every root multiplicity, is preserved by polynomial left–right equivalence and by adjoining identity variables.

Proof. The images in $k[r]\,dr$ of the two generators of R_H are

$$dH'(r) = H''(r)dr, \qquad d(rH'(r) - H(r)) = rH''(r)dr.$$

Thus $\Omega_{k[r]/R_H} \simeq k[r]/(H'')\,dr$, proving (6.2). The stable boundary theorem identifies the finite normalization morphisms on the intrinsic discriminant vertex away from the second vertex, so it identifies their relative differential modules and Fitting ideals. Under stabilization one has

$$\Omega_{(\widetilde{D}_H \times \mathbb{A}^q)/(D_H \times \mathbb{A}^q)} \simeq \mathrm{pr}_1^* \Omega_{\widetilde{D}_H/D_H}.$$

Zeroth Fitting ideals commute with this flat base change. Equality of the ideal sheaves preserves their valuations at every height-one point, hence preserves the multiplicities in (6.2), not only the reduced support. $\square$

If the support of (6.2) contains two points, the units of $k[r, \xi, \xi^{-1}, z_1, \ldots, z_q]$ and subtraction of the two corresponding principal-prime equations force the induced coordinate change to have the form $r \mapsto ar + b$, $a \neq 0$. Proposition 2.1 then says that the full effective Hessian divisor is carried affinely, with its coefficients.

The self-intersection of (6.1) is also intrinsic. In two normalization coordinates put

$$D_s(r,u) = \frac{H'(r) - H'(u)}{r - u},$$

$$D_t(r,u) = \frac{rH'(r) - H(r) - uH'(u) + H(u)}{r - u}.$$

The genuine off-diagonal closure is the scheme

$$\Gamma_H^{\mathrm{off}} = V\big((D_s, D_t) : (r-u)^\infty\big). \tag{6.3}$$

The saturation in (6.3) is necessary: on the diagonal the divided equations specialize to $H''(r)$ and $rH''(r)$ and otherwise retain isolated cusp points. The involution $(r,u) \leftrightarrow (u,r)$ is free on the ordinary off-diagonal locus; its characteristic-zero finite quotient exists in general and retains fixed diagonal limits in degenerations. On the ordinary locus that quotient records the two normalization branches paired at each node. An isomorphism of normalization morphisms preserves the fiber product, its diagonal, and the saturated complement, while stabilization takes their product with affine space. Hence this node-pairing scheme is stable. In the exact extractor, ordinary-node status is certified by the Jacobian determinant

$$\det \frac{\partial(D_s, D_t)}{\partial(r,u)}$$

being a unit on (6.3), rather than inferred only from the number of projected branch points. Passing to $\sigma = r + u, \pi = ru$ gives the scheme quotient by the free involution; the implementation verifies that its equations pull back into the saturated ordered ideal.

Finally, let

$$\mathfrak{c}_H = \operatorname{Ann}_{\mathcal{O}_{D_H}}(\nu_{H*}\mathcal{O}_{\widetilde{D}_H}/\mathcal{O}_{D_H}) \tag{6.4}$$

be the conductor. It is functorial for the same reason. On the open locus of ordinary cusps and nodes, if $P_H(r)$ cuts out all normalization branches appearing in (6.3), then the local models $k[[t^2,t^3]] \subset k[[t]]$ and $k[[x,y]]/(xy) \subset k[[x]] \oplus k[[y]]$ give

$$\mathfrak{c}_H k[r] = (H''(r)^2 P_H(r)). \tag{6.5}$$

For arbitrary plane-curve singularities, let $f_H(S,T) = 0$ be the primitive implicit equation. Plane-curve adjunction gives the uniform generator

$$\mathfrak{c}_H k[r] = \left(\frac{(\partial f_H/\partial T)(H'(r), rH'(r) - H(r))}{H''(r)} \right). \tag{6.5a}$$

This quotient is polynomial because (6.1) is the normalization. The exact extractor uses (6.5a) without an ordinary-singularity hypothesis and checks (6.5) independently whenever the cusp/node criterion holds. The invariant retains the finite conductor morphism

$$\operatorname{Spec}(k[r]/\mathfrak{c}_H k[r]) \longrightarrow \operatorname{Spec}(R_H/(R_H \cap \mathfrak{c}_H k[r])), \tag{6.6}$$

not only its upstairs support. It records the two branches over a node and the single thick branch over a cusp. Together, (6.2)–(6.6), the point at infinity, and the labeled inverse image of the intersection with the second target boundary form the decorated normalization. The full second-boundary center divisor upstairs is $\gcd(H, H')$; a point label is inferred only when its support is uniquely certified. Every part is therefore preserved under stable polynomial left–right equivalence.

For the split quartics, the boundary label on the normalization is also checked intrinsically. If zero is the unique multiple primitive root and $H(r) = h_2 r^2 + O(r^3)$, the normalized discriminant boundary chart

$$B = \frac{H'(r)}{C}, \qquad A = \frac{rH'(r) - H(r)}{cC^2}, \qquad r = Ck$$

specializes to

$$B = 2h_2 k, \qquad A = \frac{h_2}{c} k^2.$$

A finite specialization centered at $r = \rho$ forces $H(\rho) = H'(\rho) = 0$, hence $\rho = 0$. Thus the second-boundary intersection canonically marks $r = 0$; it is not an auxiliary normalization convention.

Proposition 2.2 (nonempty ordinary boundary-clean locus)**.** *For every $N \geq 4$, the normalized admissible degree-N seed space contains a nonempty open on which the discriminant has only ordinary cusps and nodes, zero is the unique multiple primitive root, and the boundary mark $r = 0$ belongs to neither the cusp nor the node-branch scheme.*

Proof. The generic discriminant theorem proves that the ordinary nodal-cuspidal open meets the normalized admissible slice in every degree. Boundary-cleanness and avoidance of the finite cusp and node-branch schemes are also open conditions. They are nonempty on the normalized slice, as witnessed by

$$H_N^{\text{split}}(W) = \frac{1}{2} W^2 (1 - W)(1 + W^{N-3}).$$

Indeed,

$$(H_N^{\text{split}})'(1) = -1, \qquad (H_N^{\text{split}})''(1) = -(N+1) \neq -2, \qquad (H_N^{\text{split}})''(0) = 1.$$

Put $P_N = 2H_N^{\mathrm{split}}/W^2 = (1-W)(1+W^{N-3})$. This polynomial is squarefree and has no zero root. At every root α of P_N,

$$\left.\frac{2(H_N^{\mathrm{split}})'(W)}{W}\right|_{W=\alpha} = \alpha P_N'(\alpha) \neq 0.$$

Thus $\gcd(H_N^{\mathrm{split}}, (H_N^{\mathrm{split}})') = W$. The nonzero second derivative at zero excludes a cusp there. If an off-diagonal point u had the same discriminant image as zero, then $H'(u) = H(u) = 0$, contradicting the gcd calculation. Hence the boundary mark is not a node branch. The normalized seed space is irreducible, so its nonempty ordinary and boundary-clean opens intersect. □

For completeness, this decoration gives an actual generic-dimension theorem, not only a parameter count. In the normalized degree-N seed space, an isomorphism preserving zero and infinity is $r \mapsto ar$. Equality of Fitting divisors then has the form $G''(r) = cH''(ar)$. Since the normalized seeds vanish together with their first derivatives at zero, integration gives

$$G(r) = \frac{c}{a^2}H(ar).$$

The condition $G(1) = 0$ forces $H(a) = 0$, leaving at most $N-2$ generic choices, and $G'(1) = -1$ determines c. Thus the decorated-normalization moduli map is generically finite and its image has dimension $N-3$.

Proposition 2.3 (generic unramifiedness and rerooting degree)**.** *For every $N \geq 4$, the decorated-normalization moduli map is generically unramified. After restriction to a nonempty open it is etale of degree $N-2$ onto the normalization of its image, and its geometric fibers are exactly the rerootings at the nonzero simple roots of the seed.*

Proof. On the squarefree boundary-clean open, retain only the Fitting divisor on $(\mathbb{P}^1; 0, \infty)$. A normalized first-order seed deformation $H + \varepsilon\dot{H}$ which is trivial in the quotient of these divisors by normalization-line scaling satisfies

$$\dot{H}''(r) = \alpha H''(r) + \beta r H'''(r). \tag{6.7}$$

The first term is the scalar ambiguity in a defining section and the second is the infinitesimal action of $r \mapsto (1+\varepsilon\beta)r$; these are all infinitesimal automorphisms because zero and infinity are marked. Twice integrating and using $\dot{H}(0) = \dot{H}'(0) = 0$ gives

$$\dot{H} = \alpha H + \beta(rH' - 2H).$$

The tangent endpoint equations now give

$$0 = \dot{H}(1) = -\beta, \qquad 0 = \dot{H}'(1) = -\alpha,$$

so $\dot{H} = 0$. The Fitting-divisor quotient, and therefore the full decorated map, is generically unramified. On the locus with trivial divisor stabilizer, the seed space and the Fitting-divisor quotient are smooth of dimension $N-3$, hence the latter map is etale after shrinking.

For the degree statement, if a is a nonzero simple root of H, put

$$\kappa_a = -\frac{1}{aH'(a)}, \qquad G_a(w) = \kappa_a H(aw).$$

Then G_a is normalized and

$$E_{G_a}(w; s, t) = \kappa_a E_H\left(aw; \frac{s}{\kappa_a a}, \frac{t}{\kappa_a}\right). \tag{6.8}$$

Thus all parts of the decoration, including the node pairing and conductor, are transported. Generically $H = w^2 P$ with P squarefree of degree $N-2$, and these $N-2$ rerootings are distinct. Conversely, the integration argument preceding the proposition shows that every point of a decorated fiber has this form. The generic degree is therefore exactly $N-2$, and the normalization of the decorated image is generically birational to the Fitting-divisor quotient. After one final shrinking they are isomorphic, which proves the asserted etaleness onto the normalized image. $\square$

Lemma 2.4 (low-pole filtration). *Let $\deg H = N$ and set $s = H'(r)$, $t = rH'(r) - H(r)$ and $R_H = k[s,t] \subset k[r]$. For the pole-order filtration at the unique point at infinity of the projective normalization,*

$$L_m(R_H) = \{f \in R_H : \mathrm{ord}_\infty(f) \geq -m\},$$

one has

$$L_{N-2} = k, \qquad L_{N-1} = k \oplus ks, \qquad L_N = k \oplus ks \oplus kt. \tag{6.9}$$

These equalities remain valid after an arbitrary field extension.

Proof. The element r is integral over $k[s]$, while $k(s,t) = k(r)$ and $[k(r) : k(s)] = N-1$. Thus the implicit equation of D_H is monic of degree $N-1$ in t, so each class has a unique representative

$$\sum_{0 \leq j < N-1} p_j(s) t^j.$$

The monomial $s^i t^j$ has pole order $i(N-1) + jN$. If two such orders are equal with $0 \leq j, j' < N-1$, reduction modulo $N-1$ gives $j = j'$ and then $i = i'$. Hence leading poles cannot cancel, and (6.9) follows. The argument is coefficient-field independent. $\square$

Remark 2.5 (repair status). *The full-cover statement below records the intended rigidity theorem and the current proof architecture, but it is not an input to the preceding decorated normalization results or to Corollary 2.8. A repair is required before this statement is promoted from repair target to an active theorem dependency.*

Theorem 2.6 (full-cover faithfulness—repair target). *Let k be a characteristic-zero field. Let $H, G \in k[W]$ be normalized degree-N seeds:*

$$H(0) = H'(0) = H(1) = 0, \quad H'(1) = -1,$$

and similarly for G. Suppose zero has multiplicity at least two, with no restriction on collisions or multiplicities among the other roots. If their full marked incidence covers, including the regular-reconstruction opens, are stably left–right equivalent, then $H = G$. Consequently,

$$F_H \sim_{\text{stable LR}} F_G \quad \Longrightarrow \quad H = G. \tag{6.10}$$

Proof. Localize away from the second target boundary and put $K = k(C)$, $s = BC$, and $t = cAC^2$. We first fix the variance convention. If the geometric target map is $\beta : Y_H \to Y_G$, its ring map is

$$\beta^* : K[s_G, t_G, z_1, \ldots, z_q] \longrightarrow K[s_H, t_H, z_1, \ldots, z_q]. \tag{6.11}$$

Thus the repository convention $G = \beta F \alpha$ uses β^{-1} when old target coordinates are expressed in the new presentation, but uses β^* in (6.11). The latter takes an irreducible equation f_G of the discriminant to a unit times f_H.

The curve D_H is geometrically irreducible by the universal-pencil theorem and is singular: H'' is nonconstant, and $d\nu_H$ vanishes at a root of H''. In particular f_H cannot lie in a one-variable coordinate subring, since geometric irreducibility would then make it linear in that coordinate and D_H a smooth coordinate line. Dryło's stable-equivalence theorem for $K[s,t]$ therefore implies that β^* preserves the plane subring $K[s,t]$.[1]

The descended plane automorphism preserves the unique infinity place. Lemma 2.4, applied in both directions, gives on D_H

$$\beta^* s_G = \lambda s_H + b, \qquad \beta^* t_G = \mu t_H + d s_H + e, \tag{6.12}$$

where $\lambda\mu \neq 0$. Every second-boundary center is a multiple primitive root and hence has $s = t = 0$; since zero supplies a center on both sides, this gives $b = e = 0$ without requiring uniqueness. Since $r = dt/ds$ on the normalization, (6.12) induces

$$r_G = \frac{\mu r_H + d}{\lambda};$$

compose with the inverse triangular linear automorphism in (6.12). The resulting plane automorphism fixes D_H pointwise. The fixed-curve theorem of Blanc–Stampfli says that a nonidentity affine-plane automorphism can fix pointwise only a curve equivalent to a line.[2] Since D_H is singular, the composition is the identity. Hence (6.12), with $b = e = 0$, holds in the entire plane ring.

Put $a = \lambda/\mu$ and $p = d/\lambda$. Restricting to the normalization and integrating gives

$$G'(r/a + p) = \lambda H'(r), \qquad G(w) = \mu H(a(w - p)). \tag{6.13}$$

The t identity makes the integration constant zero. The parameters a, p, μ lie in k: the affine map $w \mapsto a(w - p)$ carries the finite root multiset of G to that of H, and the images of the distinct roots $0, 1$ determine a, p up to a finite constant choice; the leading coefficient then determines μ. The pullback orientation is checked without convention by the identity

$$\beta^* E_G(w) = G(w) - \lambda s_H w + \mu t_H + d s_H = \mu E_H(a(w - p); s_H, t_H). \tag{6.14}$$

The universal pencil has monodromy S_N. Its degree-N incidence field corresponds to the self-normalizing subgroup S_{N-1}, so the cover has no nontrivial deck automorphism; purely transcendental stabilization adds none. Therefore the actual cover lift in (6.14) is $W_H = a(W_G - p)$.

Let $m = \operatorname{ord}_0(G) \geq 2$. The Newton polygon of $hW^m - BCW + AC^2$ shows that the zero cluster has affine sheets: both sheets when $m = 2$, and one slope-one sheet when $m \geq 3$. It maps to the H-root $-ap$ with multiplicity m. If $-ap \neq 0$, all sheets of this nonzero multiple-root cluster are polar, since $y = (W - \gamma)/C$ has leading term $(-ap)/C$. Preservation of the reconstruction open forces $p = 0$.

The affine root-one component for G now maps to the simple H-root a. Every simple extra root is polar. If $H'(a) + a \neq 0$, the coordinate $y = (W - \gamma)/C$ has a pole. If that leading pole cancels, then $\gamma \to a$ and $x \to 0$, while the numerator of the reconstruction formula for z tends to $a - 1$. Thus regularity still forces $a = 1$. Now $G = \mu H$, and $G'(1) = H'(1) = -1$ forces $\mu = 1$. $\square$

Remark 2.7. *Theorem 2.6 concerns the full incidence cover with its affine open. Proposition 2.3 is unchanged: the coarser decorated-normalization map forgets that open and remains generically etale of degree $N - 2$. The exact-double-zero boundary-clean locus is its generic special case; higher zero multiplicity and arbitrary extra-root collisions alter the compactified rerooting groupoid but introduce no new full-cover identifications.*

[1] R. Dryło, *On the stable equivalence problem for $k[x,y]$*, Colloq. Math. 124 (2011), 247–253, Theorem 1.

[2] J. Blanc and I. Stampfli, *Automorphisms of the plane preserving a curve*, 2013.

Corollary 2.8 (degreewise weighted stable moduli)**.** *For every $N \geq 4$, weighted degree-N maps contain an $(N-3)$-dimensional family of stable polynomial left–right classes.*

Proof. By Proposition 2.2, the domain of the decorated-normalization moduli map contains a nonempty open in every degree. The map is generically finite by the preceding argument and its invariant is constant on stable classes. Its stable-class image therefore has dimension $N-3$. $\square$